

Documentation of project

BUSINESS SCENARIO Business OPPORTUNITY Personal SYNOPSIS

Business scenario – Stakeholders at AAVAIL diversified their service delivery model to accommodate international customer feedback and spending habits by introducing an a la carte service model to experiment with generating new revenue. Before replacing their initial (model at company launch) entirely, stakeholders need to verify increased revenue to justify the change. Managers find it difficult to predict monthly revenue. A service that, at any point in time, will predict the revenue for the following month for both a specific country and combined revenue. It has been advised to limit your model to the ten countries with the most revenue. Managers will ultimately be the end users of this model which will predict revenues on a monthly and mid-month interval. The ability to look ahead with calculable accuracy provides them foresight which until recently would be learned after a potentially irreversible decision was made.

After discussing the scenario with stakeholders in AAVAIL, the scope and purpose of the predictive model are clear. As an employee whom AAVAIL believed in and afforded so many opportunities, the success of this project is a shared interest. With the data provided and an attention to details, the potential risks will be known and avoidable. Management explained that stability from accurate prediction capabilities will ultimately protect and benefit employees.

Data Cleaning Process Explanation

It is important to provide details about key points in the development process so that stakeholders have an idea of what decisions and influences were involved in producing the deliverable. Because the provided data is so valuable, a key focus of the cleaning process was to salvage as many incomplete entries as verifiable. The following is a walkthrough the decision-making process of correcting for missing values. This was implemented after standardizing headers, after correcting erroneous datatypes, and understanding entries to be transactional. To fix missing values, matching unique identifiers via reference map. If for example the value for country was missing, but customer_id or invoice were present, their uniqueness is only associated with one country. In a sense, it is like a reverse look up.

Data Assessment (From what is there)

Due to variations in labeling the first step taken was to apply a list of headers. The standardized headers were: country, customer_id, invoice, price, stream_id, times_viewed, year, month, day. After cleaning and preparations for modeling, analysis began with focus on monthly changes, observed patterns, and the resulting explorations. Rules were developed from observation for guiding conditions for things that were not explicitly stated such as: USA tiered model should only have one price, whereas the service transaction model relies on times_viewed and price, to calculate. And between the two of those if the same stream_id is found then that can be to assess the difference in revenue and support the stakeholder decision to switch all markets over to a la carte. Further that country translates to market boundary.

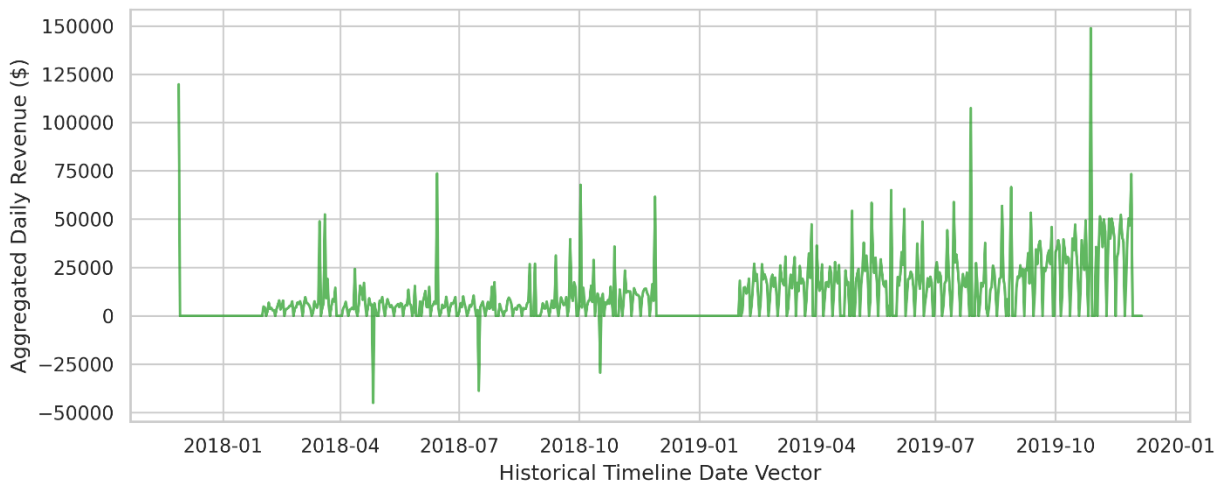
Hypothesis

The difference between the tiered subscription and a la carte will be most clearly observed by the singular price of USA, and the price multiplied by time_viewed for all other countries other than USA. Secondary hypothesis or supporting observation is that the subscription-based model is limited by the number of customers it has. Whereas the a la carte model allows for limitless transactions by the customer should they choose. Even if the price is reduced per transaction the fact that one customer can have multiple transactions will generate more income overtime

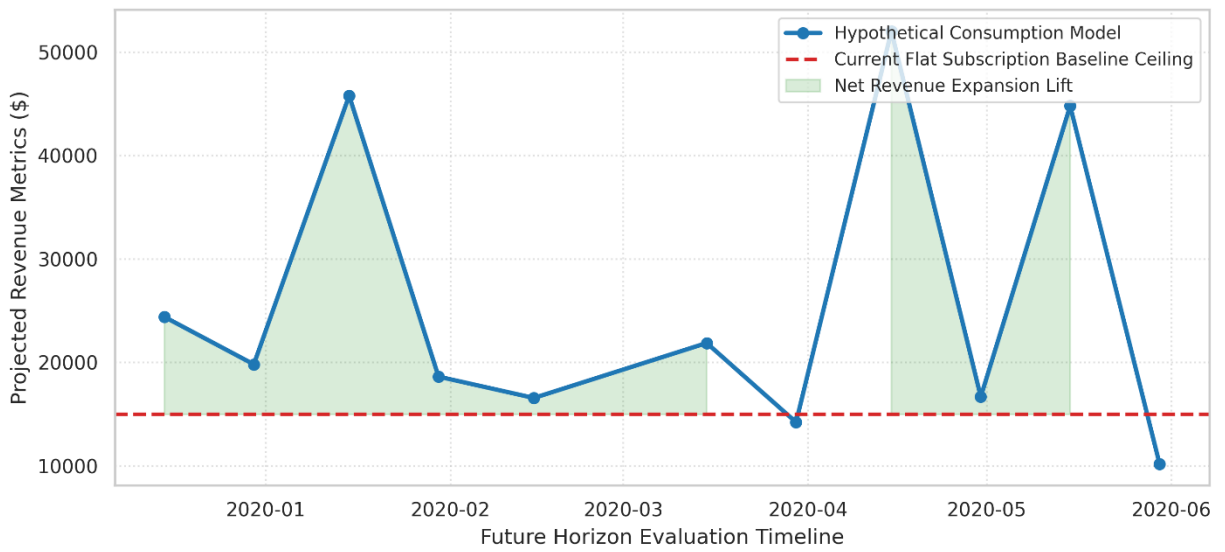
Proof of success

Forecast Date	Milestone	Projected Revenue
:-----	:-----	:-----
2019-12-15	Day 15	\$24,413.26
2019-12-30	Day 30	\$19,831.38
2020-01-15	Day 15	\$45,794.52
2020-01-30	Day 30	\$18,644.34
2020-02-15	Day 15	\$16,582.76
2020-03-15	Day 15	\$21,892.54
2020-03-30	Day 30	\$14,237.34
2020-04-15	Day 15	\$52,012.22
2020-04-30	Day 30	\$16,694.13
2020-05-15	Day 15	\$44,797.36
2020-05-30	Day 30	\$10,222.89

Historical Ingested Timeline: Cleaning Verification Loop



Pricing Strategy Evaluation: Consumption Forecasting vs. Subscription Ceiling



Feature Seasonality Analysis: Projected Average Revenue by Operational Day

